Comparative Performance Analysis of Pandas, Pandas Chunking, and Dask Using the Avazu CTR Prediction Dataset

Student Name

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Abstract

This report compares three methods for processing a large dataset: Standard Pandas, Pandas Chunking, and Dask DataFrame. The evaluation focuses on execution time, memory consumption, and the consistency of the obtained results. The Avazu Click-Through Rate (CTR) Prediction Dataset is used for this experiment.

1 Introduction

Processing large datasets can require significant memory and computational resources. Different data-processing tools provide different solutions to these challenges.

The objective of this practical work is to compare Standard Pandas, Pandas Chunking, and Dask in terms of execution time, memory consumption, and accuracy when processing a large CSV file.

2 Dataset Description

The dataset used in this experiment is the **Avazu Click-Through Rate (CTR) Prediction Dataset**. The input file is named `train.csv` and contains 40,428,967 rows.

The main variable used in this experiment is `click`, which indicates whether an advertisement was clicked.

3 Methodology

3.1 Task 1: Standard Pandas

Standard Pandas reads the entire CSV file into memory using the `read_csv()` function. The program calculates the total number of rows, the total number of clicks, and the click

rate.

This method is simple to implement, but loading the complete dataset can require a large amount of RAM.

3.2 Task 2: Pandas Chunking

Pandas Chunking processes the dataset in smaller portions using the `chunksize` parameter. In this experiment, each chunk contains 100,000 rows.

The program processes one chunk at a time and accumulates the number of rows and clicks. This approach reduces memory consumption because the entire dataset is not loaded at once.

3.3 Task 3: Dask DataFrame

Dask DataFrame divides the dataset into partitions and builds a task graph for the requested operations. It uses lazy evaluation, so computations are performed when a result is requested using functions such as `compute()`.

This approach can improve performance and facilitate the processing of large datasets.

4 Results and Comparison

The click rate is calculated as follows:

$$ClickRate = \frac{TotalClicks}{TotalRows} \times 100$$

Table 1 presents the experimental results.

Table 1: Performance comparison of the three methods

Method	Time (s)	Memory (GB)	Rows	Clicks	Rate
Standard Pandas	370.66	6.96	40,428,967	6,865,066	16.98%
Pandas Chunking	344.85	0.16	40,428,967	6,865,066	16.98%
Dask	245.71	0.38	40,428,967	6,865,066	16.98%

4.1 Execution Time

Dask achieved the shortest execution time, completing the task in 245.71 seconds. It was approximately 33.7% faster than Standard Pandas. Pandas Chunking achieved a smaller improvement of approximately 7.0%.

4.2 Memory Consumption

Pandas Chunking had the lowest reported memory consumption, at 0.16 GB. Dask used 0.38 GB, whereas Standard Pandas used 6.96 GB. These results demonstrate the memory advantage of processing data in smaller portions or partitions.

4.3 Result Consistency

All three methods returned the same number of rows, total clicks, and click rate. The results were 40,428,967 rows, 6,865,066 clicks, and a click rate of 16.98%.

This agreement indicates consistent results for the statistics measured in this experiment.

5 Discussion

The results show that each method has different advantages. Standard Pandas is straightforward but requires substantial memory. Pandas Chunking is suitable for systems with limited RAM. Dask achieved the best execution time in this experiment while maintaining relatively low memory consumption.

These results should be interpreted in the context of the experimental setup, because execution time and memory usage can vary depending on hardware, configuration, and measurement method.

6 Conclusion

This practical work compared three approaches for processing the Avazu CTR Prediction Dataset. Dask was the fastest method, while Pandas Chunking was the most memory-efficient according to the recorded measurements. All three approaches produced consistent results for the evaluated statistics.

Therefore, Dask is a suitable choice when processing speed is important, whereas Pandas Chunking is a practical solution when memory resources are limited.